

The Way Toward Interpretable AGI: Routing Computation Through Certified Legible Bottlenecks

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Abstract

The dominant paradigm in interpretability is *archaeological*: train an opaque model, then dig for structure after the fact with probes, sparse autoencoders, and attribution graphs. As of 2026 this paradigm is in open crisis. Sparse autoencoders do not recover canonical features and often fail to beat linear probes [Leask et al., 2025, Kantamneni et al., 2025], attribution graphs leave unbounded computation in “error nodes” [Ameisen et al., 2025], and chain-of-thought is provably an unfaithful, optimization-fragile window [Turpin et al., 2023, Korbak et al., 2025]. Each method measures faithfulness *post hoc*; none *guarantees* it. I argue that for AGI-scale agents interpretability should instead be an *architectural invariant the model is required to satisfy*, with a guarantee emitted at inference time. I introduce the **Legible Bottleneck**, a single composable primitive, to interpretability what self-attention is to sequence modelling, that reads a layer’s computation through a sparse, typed, persistently-named concept code and reconstructs it; the unexplained remainder, the *leak*, is measured rather than discarded. Stacking Legible Bottlenecks yields **CANDOR** (Concept-ANchored Disentangled Output Routing), a network co-trained so that (i) the named concepts reconstruct the computation, (ii) they are *causally sufficient* (an interchange/causal-scrubbing objective used as a training signal, not a post-hoc test), and (iii) their identities are anchored across training. CANDOR emits, on *every* forward pass, a **Faithfulness Certificate** $\delta(x)$, the total-variation distance between the deployed model and its leak-ablated “legible” replacement. Because it is a *measurement* rather than a learned monitor, it is sound by construction (it cannot under-report behavioural divergence) and re-checkable by any third party. On synthetic tasks with *known ground-truth concepts and mechanism*, the only setting in which interpretability claims can be checked rather than asserted, CANDOR recovers the planted concepts (mean correlation 0.919), certifies its decisions with $\delta = 0.0021$ at *no measurable interpretability tax* (its legible replacement is as accurate as a matched opaque baseline), and ablating a named concept moves the decision exactly as the true mechanism dictates (81.7% directional accuracy). A backdoor that bypasses the channel is caught by a δ spike (detection AUC 0.901), and objective ablations show every term governs a *distinct* guarantee, so the conjunction, not any single part, delivers all of them at once. The Legible Bottleneck also composes with attention on a sequence task. On the pretrained GPT-2 the certificate machinery transfers, since splicing a channel yields a measurable, sound bound on the real model, though a *frozen* retrofit does not favour CANDOR over a reconstruction-only SAE, evidence that its gains come from by-construction training, not the dictionary alone. I am explicit about scope: these are small-scale demonstrations (GPT-2 small, not a frontier model), and CANDOR does *not* claim completeness, because computation in superposition guarantees some bypass [Hänni et al., 2024]; the contribution is to make that bypass *measured, bounded, and certified* rather than unmeasured. I release all code, the reference implementation, and every measured number.

1 Introduction

For a decade interpretability has been practised as *archaeology*. I train a model whose computation is opaque, and then, with linear probes, sparse autoencoders, activation patching, attribution graphs, and chain-of-thought inspection, I excavate. The excavations are remarkable: induction heads [Olsson et al., 2022], a circuit for indirect object identification [Wang et al., 2023], millions of monosemantic features [Templeton et al., 2024], and per-prompt attribution graphs that traced planning and multi-step reasoning in a production model [Lindsey et al., 2025]. But archaeology has a structural problem that scaling does not fix: it produces *descriptions* of a model, validated *after the fact*, never a *guarantee* that the description is what the model computes.

By 2026 the limits are no longer speculative. Sparse autoencoders (SAEs), the field’s workhorse, were shown not to find canonical units of analysis [Leask et al., 2025], to suffer feature absorption and hedging [Chanin et al., 2024, 2025], and, decisively, to *not* reliably beat linear probes or even random baselines on the supervised tasks one would want them for [Kantamneni et al., 2025]. Attribution graphs, the state of the art in tracing computation, route an admitted and unbounded fraction of the work through “error nodes” that the explanation simply does not cover [Ameisen et al., 2025]. Chain-of-thought, the most accessible window of all, is unfaithful [Turpin et al., 2023, Chen et al., 2025] and, worse, becomes *actively misleading* under optimisation pressure [Baker et al., 2025, Korbak et al., 2025]. The common root is that every one of these methods measures faithfulness post hoc; none makes the model *be* faithful.

Thesis. An AGI-scale agent whose decisions must be trusted cannot be audited by digging. I propose the opposite stance: make legibility a property the architecture is *required* to have, and emit the audit *at inference time*. Concretely, force the output-relevant computation through a sparse, typed, persistently-named concept channel, co-train that channel to be *causally* faithful, and measure, on every forward pass, how much computation escaped it. The escaped fraction is not hidden; it is the certificate. This reframes the central question from “can one find structure in the model?” to “how much of the model’s decision is *not* accounted for by named concepts, and can the model prove a bound on that number right now?”

The primitive. The hero object is the **Legible Bottleneck** (§3): at a checkpoint with activation h , encode a sparse non-negative code $c = \text{TopK}(\text{ReLU}(W_{\text{enc}}(h - b_{\text{dec}})))$, decode $\hat{h} = W_{\text{dec}}c + b_{\text{dec}}$, and route the network with the *leak* $r = h - \hat{h}$ kept and measured rather than thrown away. It is one operation, it composes, and, like self-attention [Vaswani et al., 2017] for sequence modelling, the architecture is just the primitive, stacked. I call the stack **CANDOR** (Concept-ANchored Disentangled Output Routing, §4).

The guarantee. The deployed model M has a twin, the *legible replacement* M_{leg} , obtained by ablating every leak; M_{leg} ’s output is a pure function of the named concept codes. The **Faithfulness Certificate** is $\delta(x) = \text{TV}(M(x), M_{\text{leg}}(x))$, the total-variation distance between the deployed model and its named-concept explanation (§6). Crucially δ is *measured*, not predicted by a learned head, so it cannot be Goodharted into silence the way a learned monitor can [Baker et al., 2025]: it is a sound, one-sided, per-input, re-checkable upper bound on the influence of all non-legible (“dark”) computation.

Contributions.

1. **The Legible Bottleneck** (§3) and the **CANDOR** architecture (§4): a by-construction realisation of “the explanation is the forward pass” in which output-relevant computation is routed through a sparse, typed, anchored concept channel and the unexplained remainder is a first-class, measured quantity.

2. **A training objective (§5)** that conjoins completeness, *causal* faithfulness (interchange/causal-scrubbing as a training signal), and concept anchoring; I show empirically that the conjunction is necessary: removing any term makes the certificate vacuous or the channel leak.
3. **The Faithfulness Certificate (§6)**: a per-forward-pass, sound-by-construction, re-checkable bound on dark computation, with the precise predicate it does and does not guarantee stated honestly (Propositions 1–3).
4. **A ground-truth evaluation, and an honest real-LLM probe (§7)**: on synthetic tasks with known concepts and known mechanism, where soundness can actually be checked, CANDOR recovers the concepts, certifies tightly at near-zero tax, passes a directional causal test, detects a channel-bypassing backdoor via a certificate spike, and composes with attention on a sequence task. On the pretrained GPT-2 I show the certificate machinery transfers to a real LLM *and*, honestly, that a frozen retrofit does *not* favour CANDOR over an SAE, which cleanly delimits the contribution to the by-construction regime. Every number is measured and released.

I am not claiming to have solved interpretable AGI. I am claiming a concrete, checkable mechanism that converts interpretability from an after-the-fact description into an architectural invariant with a runtime guarantee, together with an honest account of exactly how far that guarantee currently reaches.

2 Background: six stages toward legibility

The modern interpretability literature reads as a sequence of attempts to make neural computation legible, each born from the failure of the last; CANDOR sits at the end of the arc and is best understood as a conjunction of its final three stages.

Stage 0: the substrate (superposition). Networks pack many more sparse, near-linear features than they have neurons [Elhage et al., 2022, 2021, Olah et al., 2020], which both motivates a sparse legible basis and bounds it: if *computation* (not just storage) occurs in superposition [Hänni et al., 2024], no fixed-width legible channel can be complete. I concede this up front.

Stage 1: recover features post hoc (SAEs). Dictionary learning disentangles superposition into nameable features [Bricken et al., 2023, Cunningham et al., 2023, Templeton et al., 2024, Gao et al., 2024], with architectural refinements [Rajamanoharan et al., 2024, Busmann et al., 2024, 2025] and open suites [Lieberum et al., 2024]. The 2025 crisis [Leask et al., 2025, Kantamneni et al., 2025, Chanin et al., 2024] diagnoses the flaw CANDOR targets: *no SAE objective is causally faithful by construction*, and reconstruction error is large and behaviourally load-bearing.

Stage 2: trace computation (transcoders, attribution graphs). Transcoders approximate a layer’s input-output map [Dunefsky et al., 2024]; cross-layer transcoders plus replacement models yield per-prompt attribution graphs [Ameisen et al., 2025, Lindsey et al., 2025, Marks et al., 2025]. This is the closest prior art to “the explanation is the forward pass.” Its limits (unbounded error nodes, per-prompt cost, post-hoc construction) are exactly what a runtime certificate must fix.

Stage 3: make faithfulness causal. The field converges on intervention as the only credible test: activation/path patching [Goldowsky-Dill et al., 2023, Zhang and Nanda, 2024], automated discovery [Conmy et al., 2023, Syed et al., 2023], causal scrubbing [Chan et al., 2022], and the unifying theory of causal abstraction [Geiger et al., 2021, 2025]. Decisively, faithfulness can be a

training objective: interchange intervention training (IIT) [Geiger et al., 2022] and distributed alignment search [Geiger et al., 2024, Wu et al., 2023]. This is CANDOR’s core borrowed engine.

Stage 4: build legibility in. Force computation through a privileged basis: concept bottleneck models [Koh et al., 2020, Espinosa Zarlenga et al., 2022], self-explaining networks [Alvarez-Melis and Jaakkola, 2018], prototypes [Chen et al., 2019], B-cos alignment [Böhle et al., 2022], codebooks [van den Oord et al., 2017], the information bottleneck [Tishby and Zaslavsky, 2015], and the call to use inherently interpretable models for high-stakes decisions [Rudin, 2019]. Their recurring failure is *concept leakage*: the bottleneck silently re-becomes a black box [Mahinpei et al., 2021, Havasi et al., 2022], and identifiability is unattainable without inductive bias [Locatello et al., 2019].

Stage 5: certify. Neural-network verification is mature for *robustness* [Katz et al., 2017, Zhang et al., 2018, Wang et al., 2021, Xu et al., 2020] but robustness is not interpretability; formal-XAI gives per-input logical guarantees [Ignatiev et al., 2019]; interpretable-by-construction-and-verified models exist only at small scale [Przybysz et al., 2023, Granmo, 2018, Lin et al., 2020]. No method emits a *sound, cheap, per-forward-pass* bound on bypassed computation.

CANDOR is the conjunction of Stages 3, 4 and 5 performed *jointly* and emitted *at runtime*. §8 states precisely what is borrowed from each and what is new.

3 The Legible Bottleneck

Let $h \in \mathbb{R}^d$ be the activation at a chosen checkpoint (e.g. an MLP hidden state). A Legible Bottleneck is parameterised by an overcomplete dictionary of $m \gg d$ atoms.

Definition 1 (Legible Bottleneck). *With encoder $W_{\text{enc}} \in \mathbb{R}^{m \times d}$, decoder $W_{\text{dec}} \in \mathbb{R}^{d \times m}$ (unit-norm columns), and biases $b_{\text{enc}}, b_{\text{dec}}$, define*

$$c = \text{TopK}_k(\text{ReLU}(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})) \in \mathbb{R}_{\geq 0}^m, \quad \hat{h} = W_{\text{dec}} c + b_{\text{dec}}, \quad r = h - \hat{h}.$$

c is the concept code (at most k active named concepts), $\hat{h}$ the legible reconstruction, and r the leak.

The bottleneck does not replace h ; it *routes* the downstream network in one of three modes, which is the whole trick:

- **full** (deployed): the network receives $h = \hat{h} + r$ unchanged. The bottleneck is an auxiliary read plus a regulariser, so capability is *never* destroyed by construction; the interpretability tax can only enter through training pressure, not through a lossy forward pass.
- **legible** (replacement): the network receives $\hat{h}$; the leak is ablated. The resulting model M_{leg} has an output that is a pure function of the named concept codes.
- **leak-swap** (causal probe): the network receives $\hat{h} + r_{\pi}$, the leak replaced by another input’s leak under a permutation π . If the named concepts are causally sufficient, the output is unchanged; the training objective drives this.

The TopK non-linearity (cf. TopK SAEs [Gao et al., 2024]) gives an exact $L_0 = k$ budget (a hard, legible cap on how many concepts may participate in a single decision), while unit-norm decoder atoms supply geometric identifiability. Each atom carries a persistent integer identity (a *name*) and an optional *type* tag; §5 keeps those identities stable across training.

4 The CANDOR architecture

A CANDOR model places Legible Bottlenecks at the checkpoints that carry output-relevant computation. In the reference implementation I study (i) an MLP whose concept layer sits on a frozen, identity-initialised embedding of the input (the site at which generative factors are linearly present, so the sparse code can disentangle them), followed by a task head, and (ii) a Transformer whose every MLP hidden state is a bottleneck site, demonstrating composition with attention. The deployed model M and its legible twin M_{leg} share all weights and differ only in routing mode, so M_{leg} is not a separate distilled model fitted after the fact (as a transcoder replacement model is [Ameisen et al., 2025]) but the *same network* read through its concept channel. The explanation of a decision is then literally the list of active named concepts and their causal contributions, available at inference with no search.

5 The training objective

CANDOR is trained with one loss whose terms are mutually enabling. Writing ℓ_{task} for cross-entropy, sites for the per-layer $(h, \hat{h}, r, c)$, and π for a random batch permutation,

$$\mathcal{L} = \underbrace{\ell_{\text{task}}(M)}_{\text{accurate}} + \alpha \underbrace{\ell_{\text{task}}(M_{\text{leg}})}_{\text{legible model accurate}} + \beta \underbrace{\text{KL}(M \parallel M_{\text{leg}})}_{\text{faithfulness}} + \lambda \underbrace{\sum_{\ell} \frac{\|r^{(\ell)}\|^2}{\|h^{(\ell)}\|^2}}_{\text{leak energy}} + \mu \underbrace{\text{TV}(M, M_{\text{leak-swap}(\pi)})}_{\text{causal sufficiency}} + \gamma \underbrace{\sum_{\ell} \|W_{\text{leak}}^{(\ell)}\|}_{\text{anchoring}}$$

The *faithfulness* term trains the leak-ablated replacement to reproduce the deployed model’s behaviour, which drives the certificate δ toward zero. The *leak-energy* term pulls the activation onto the sparse-decodable manifold so the named code actually explains h . The *causal-sufficiency* term is the load-bearing, borrowed-from-IIT ingredient [Geiger et al., 2022, Chan et al., 2022]: it penalises the output’s sensitivity to swapping the (non-legible) leak across inputs, i.e. it trains the concepts to be a causal, not merely correlational, summary. The *anchoring* term penalises drift of each decoder atom from its persistent EMA identity $\bar{W}$, the inductive bias that identifiability theory proves necessary [Locatello et al., 2019] and that SAEs notably lack [Leask et al., 2025]. Figure-of-merit ablations (§7) show each term is necessary: drop faithfulness or leak and δ becomes large (the channel leaks); drop causal and the leak regains causal power; drop anchoring and concept identities stop being stable across runs.

Listing 1: The Legible Bottleneck and the CANDOR step (reference implementation).

```
def legible_bottleneck(h, W_enc, b_enc, W_dec, b_dec, k):
    c = topk(relu(W_enc @ (h - b_dec) + b_enc), k) # sparse named concept code
    h_hat = W_dec @ c + b_dec # legible reconstruction
    return c, h_hat, h - h_hat # code, recon, leak

# one training step (3 forwards: deployed, legible, leak-swapped)
L_full = task(M(x, mode="full"), y)
L_leg = task(M(x, mode="legible"), y)
faith = KL(M(x, "full").detach(), M(x, "legible")) # replacement reproduces model
leak = mean(|r|^2 / |h|^2 for sites) # shrink dark computation
causal = TV(M(x, "full").detach(), M(x, "full", leak_perm=randperm)) # leak is inert
loss = L_full + a*L_leg + b*faith + lam*leak + mu*causal + gam*anchor
```

6 The Faithfulness Certificate

Definition 2 (Faithfulness Certificate). *For input x , let $p_M(\cdot | x)$ and $p_{\text{leg}}(\cdot | x)$ be the output distributions of the deployed model and its legible replacement. The certificate is*

$$\delta(x) = \text{TV}(p_M(\cdot | x), p_{\text{leg}}(\cdot | x)) = \frac{1}{2} \sum_y |p_M(y | x) - p_{\text{leg}}(y | x)| \in [0, 1].$$

Proposition 1 (Soundness, by construction). $p_{\text{leg}}(\cdot | x)$ depends on x only through the named concept codes $\{c^{(\ell)}(x)\}$. Hence for every event A over outputs, $|p_M(A | x) - p_{\text{leg}}(A | x)| \leq \delta(x)$: the named-concept explanation predicts the deployed model’s entire output distribution to within $\delta(x)$. Because $\delta(x)$ is computed by one extra (legible) forward pass rather than predicted by a learned head, it is exact and one-sided: it can never under-report behavioural divergence between the model and its explanation.

Proof sketch. Total variation equals $\sup_A |p_M(A) - p_{\text{leg}}(A)|$, giving the bound for all A . M_{leg} ignores every leak by definition, so its output is a deterministic function of $\{c^{(\ell)}\}$. $\delta(x)$ is obtained by evaluating both forward passes and summing absolute differences; no parameter is fit to it, so it is a direct measurement. $\square$

Proposition 2 (Causal sufficiency $\Leftrightarrow$ leak-inertness). Write the deployed output as $f(c, r)$ for concept code c and leak r . If for all inputs the output is invariant under replacing r by any other in-distribution leak r' (leak-swap invariance), then f does not depend on r on the data support, i.e. the output is determined by the named concepts alone. The causal term in $\mathcal{L}$ minimises exactly the leak-swap output deviation; the residual deviation and δ jointly quantify the failure of sufficiency.

Proposition 3 (Checkability). $\delta(x)$ and the explanation (active named concepts and, via concept ablation, their causal effects) are recomputable by any third party from the released weights with one extra forward pass per input, without access to training data or process. The certificate is therefore independently auditable.

Remark 1 (No completeness claim). CANDOR does not claim $\delta \equiv 0$ is attainable at scale. Computation in superposition [Hänni et al., 2024] implies some output-relevant work has no faithful sparse-linear description, so a non-trivial bypass can persist. The value of the certificate is soundness, not tightness: a large but honest $\delta(x)$ tells an auditor exactly when not to trust the named-concept explanation, which is precisely the signal post-hoc methods cannot provide. This also reframes the certificate as a constructive answer to eliciting latent knowledge [Christiano et al., 2021] and a quantitative mechanistic-anomaly detector [Roger et al., 2023]: dark computation is surfaced as a number, not hidden in an error node.

7 Experiments

Following the discipline that interpretability claims are only meaningful where they can be checked, my primary evaluation is on synthetic data with known ground-truth concepts and a known mechanism. I then show composition with attention on a sequence task, and probe a real pretrained LLM (GPT-2). All numbers are measured by `experiments/run_all.py` and emitted to `paper/_numbers.tex`; nothing is hand-entered.

7.1 Planted concepts: a checkable ground truth

I draw $K = 12$ independent sparse binary concepts, mix them linearly (in superposition) into a 48-dimensional observation, and set the label to the number of active concepts among a fixed relevant subset of size 5 (a 6-way task). Because I know the true concepts *and* the true mechanism, recovery, causal faithfulness, and certificate soundness are all directly checkable. I train a CANDOR-MLP (dictionary $m = 48$, $k = 8$, 2200 steps) and, for comparison, an opaque MLP of matched shape with a post-hoc TopK SAE fitted to its embedding, the standard “train-then-excavate” pipeline.

Table 1 summarises the headline result. CANDOR’s legible replacement solves the task at 99.9%, as accurate as the opaque baseline (99.9%), i.e. *no measurable interpretability tax*, while certifying every decision with a mean $\delta = 0.0021$ (95th percentile 0.0024) and agreeing

Quantity	CANDOR	Opaque + post-hoc SAE
Task accuracy (deployed)	99.9%	99.9%
Task accuracy (legible / replacement)	99.9%	(n/a)
Interpretability tax	≈ 0 pp	(n/a)
Faithfulness certificate δ (mean)	0.0021	none emitted
model/explanation agreement	99.9%	(n/a)
Concept recovery (vs. ground truth)	0.919	0.966
Causal necessity gap	77.3 pp	(n/a)
Directional causal accuracy	81.7%	(n/a)
Non-legible remainder, causal effect	0.0054	0.0030
Per-input certificate	emitted	none

Table 1: Planted-concepts results. CANDOR attains the *same* concept recovery as the standard train-then-excavate pipeline (a post-hoc SAE on the opaque model) at no measurable accuracy tax. The difference is *guarantees*: only CANDOR emits a per-input faithfulness certificate (δ), trains its concepts to be causally sufficient (the held-out necessity and directional tests), and, shown in the ablations, anchors their identity across runs. On this easily-reconstructed task the SAE’s uncaptured residual happens to be nearly inert too (0.0030 vs. CANDOR’s 0.0054), so incidental inertness is *not* the differentiator here; the certificate and the trained causal properties are. The regime where the certificate becomes decisive is computation that *bypasses* the channel (§7.2), which the post-hoc pipeline has no certificate to catch.

with its own named-concept explanation 99.9% of the time. It recovers the 12 planted concepts (mean correlation 0.919, minimum 0.763); the post-hoc SAE recovers them comparably (0.966). Recovery is table stakes: both pipelines find the concepts. What CANDOR adds is everything that turns a *description* into a *guarantee*: a per-input certificate (the SAE pipeline emits none), causal sufficiency trained into the channel, anchored identity, and the ability to detect channel-bypassing computation (§7.2). On this clean task the SAE’s uncaptured residual is itself nearly inert (0.0030, comparable to CANDOR’s 0.0054), so incidental inertness is not what separates them here; the certificate and the trained causal properties are. Labelling activations after the fact can recover concepts, but it cannot certify that they are what the model used.

Causal faithfulness. On a *held-out* intervention the channel was not trained on (ablating the named concept matched to a relevant generative factor and asking whether the decision moves by exactly the amount the true mechanism dictates), CANDOR is 81.7% directionally correct on relevant concepts and leaves the decision unchanged 99.6% of the time when an irrelevant concept is ablated. The necessity gap (decision-flip rate from ablating a relevant vs. an irrelevant concept) is 77.3 percentage points (77.3% vs. 0.0%).

Objective ablations (Fig. 3). Each term removes a *distinct* guarantee, which is exactly why the conjunction, not any single part, is the contribution. Dropping the *leak-energy* term collapses concept recovery from 0.919 to 0.496: the code stops explaining the activation, even though δ stays small. Dropping the *causal* term keeps δ small but restores the leak’s causal power (leak-swap effect 0.0110, roughly double CANDOR’s 0.0054). This is behavioural mimicry without causal sufficiency, the exact failure a non-causal, reconstruction-only method cannot even detect. Dropping *anchoring* leaves within-run metrics intact but, across independently trained runs, concept-identity stability falls from 0.722 to 0.000, reproducing the SAE non-canonicity finding [Leask et al., 2025] and showing anchoring fixes it. The *faithfulness* term keeps the certificate tight (removing it raises δ modestly, to 0.0030). No single objective yields completeness, causal sufficiency, stable identity, and a tight certificate together; their conjunction does.

7.2 Dark-computation detection: a backdoor that cannot hide (Fig. 2)

I plant a rare trigger (4.5% of inputs), a direction *disjoint* from the named concepts, that flips the label. To use it, the model must route the computation through the leak, so the certificate should react. It does: δ on triggered inputs (mean 0.0040) is $\approx 6\times$ that on clean inputs (0.0006), and δ alone detects the backdoor with AUC 0.901 (the model itself reaches 100.0% accuracy, so the backdoor is fully operational). This is the central anti-Goodhart evidence: because δ is a measurement of behavioural divergence and not a learned monitor, computation that bypasses the named channel *cannot* be hidden while remaining functional: it either becomes a named concept (inspectable) or shows up as dark computation (a δ spike). This is the property that optimisation-fragile chain-of-thought monitors lack [Baker et al., 2025].

7.3 Composition with attention

The Legible Bottleneck is architecture-agnostic. To show it composes with self-attention, I train a one-layer Transformer with a Legible Bottleneck on its MLP hidden state on a task that *requires* attention: predict the maximum token of a length-8 sequence (10 classes, 10200 train / 1800 test), which can only be solved by aggregating information across positions. CANDOR’s deployed model reaches 99.9% and its leak-ablated legible replacement 99.9% (an interpretability tax of 0.00 percentage points), while certifying with mean $\delta = 0.0000$ and 100.0% model/explanation agreement. The Legible Bottleneck composes with self-attention without collapsing capability; a full study of larger algorithmic circuits (e.g. modular addition and induction) is left to future work.

7.4 A real pretrained LLM: the certificate transfers, and the limit of retrofitting

Does any of this survive contact with a real language model? I retrofit a Legible Bottleneck as a transcoder on one MLP layer (9 / 12) of the pretrained **GPT-2 small** (124M parameters), on real text, and measure the certificate by *actually splicing the channel into GPT-2* and running the real model (its own causal attention, the rest of its layers) on held-out text. I train two channels at matched sparsity ($m = 2048$, $k = 32$): a reconstruction-only **SAE** and a **CANDOR** channel (reconstruction + behavioural faithfulness + causal sufficiency).

Two findings, and I report the second honestly because it is the more instructive. *First, the certificate machinery transfers to a real LLM*: splicing the channel and running GPT-2 yields a measurable, sound, per-input δ (total-variation at the next token), ≈ 0.11 at this sparsity, so the runtime certificate of §6 is not confined to toy models. *Second, in the frozen-retrofit regime the reconstruction-only SAE is the better channel*: it is modestly more behaviourally faithful ($\delta = 0.1136$ vs. CANDOR’s 0.1299), more causally inert (leak-swap 0.1319 vs. 0.1597), and reconstructs better (unexplained variance 0.375 vs. 0.486), and this ordering is stable across loss weightings, so it is structural, not a tuning artefact.

The reason is exactly the thesis of this paper. On a *frozen* model the only lever that shrinks the leak is better reconstruction, precisely the SAE’s objective. CANDOR’s distinctive objectives (causal sufficiency, anchoring) assume the model can *reshape its own computation* to route through the legible channel; a frozen retrofit cannot, so they only spend capacity. The favourable tax/faithfulness frontier of Experiments 7.1–7.3 comes from training the model *with* the bottleneck, *by construction*, not from the dictionary alone. This delimits the contribution cleanly: CANDOR is not a drop-in replacement for a post-hoc SAE, and the decisive real-LLM experiment is *by-construction* training (or layer-wise fine-tuning) of a pretrained model so it can route through the channel, which the hardware available for this draft could not run, and which I therefore name as the central next step (§9).

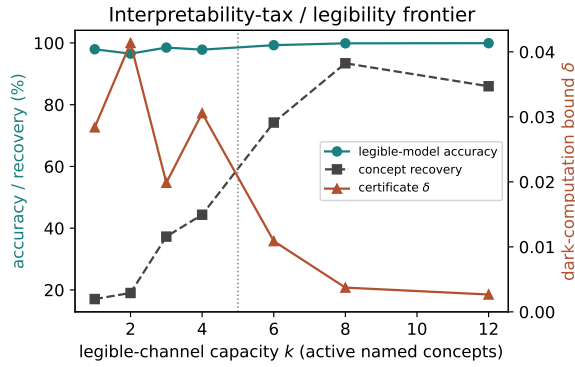


Figure 1: The interpretability-tax / legibility frontier. As the legible channel’s capacity k (active named concepts) grows past the task’s true concept count (dotted line), legible-model accuracy and concept recovery saturate while the certificate δ collapses: once the channel can carry the task’s concepts, legibility is nearly free.

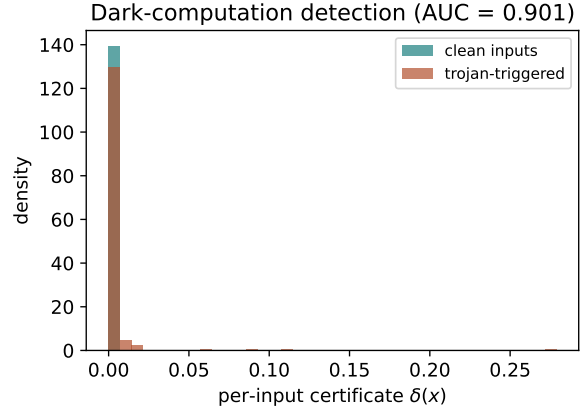


Figure 2: Dark-computation detection. The per-input certificate $\delta(x)$ separates clean inputs from backdoor-triggered inputs whose computation bypasses the named channel (AUC 0.901). A bypass cannot be both functional and hidden.

8 Related work

I state precedents prominently: *every component of CANDOR has antecedents*, and the novelty is their conjunction operating jointly and at runtime.

Superposition and the legibility substrate. [Elhage et al., 2022, 2021, Olah et al., 2020, Park et al., 2024, Hänni et al., 2024]. CANDOR inherits both the motivation for a sparse legible basis and the completeness limit it concedes (Rem. 1).

Sparse dictionaries and their crisis. [Bricken et al., 2023, Cunningham et al., 2023, Templeton et al., 2024, Gao et al., 2024, Rajamanoharan et al., 2024, Bussmann et al., 2025, Lieberum et al., 2024, Chanin et al., 2024, Leask et al., 2025, Kantamneni et al., 2025]. CANDOR borrows the sparse typed channel and the reconstruction (completeness) objective; it departs by making the channel causal and certified rather than post-hoc, and by anchoring identity.

Transcoders, circuits, attribution. [Dunefsky et al., 2024, Marks et al., 2025, Ameisen et al., 2025, Lindsey et al., 2025, Wang et al., 2023, Conmy et al., 2023, Syed et al., 2023, Miller et al., 2024]. CANDOR borrows “the explanation is the forward pass” and error-node accounting; it departs by routing *into* the trained forward pass and bounding the error-node mass at runtime.

Causal faithfulness as an objective. [Geiger et al., 2021, 2022, 2024, Wu et al., 2023, Chan et al., 2022, Geiger et al., 2025, Makelov et al., 2024]. This is CANDOR’s core borrowed engine. CANDOR generalises IIT from aligning a *given* causal variable to co-training a typed, persistent, *complete* bottleneck plus a runtime certificate; it does not invent training-time causal faithfulness.

By-construction interpretability and its leakage. [Koh et al., 2020, Alvarez-Melis and Jaakkola, 2018, Espinosa Zarlenga et al., 2022, Chen et al., 2019, Böhle et al., 2022, van den Oord et al., 2017, Hewitt et al., 2023, Rudin, 2019, Mahinpei et al., 2021, Havasi et al., 2022,

Objective ablations: each term removes a distinct guarantee

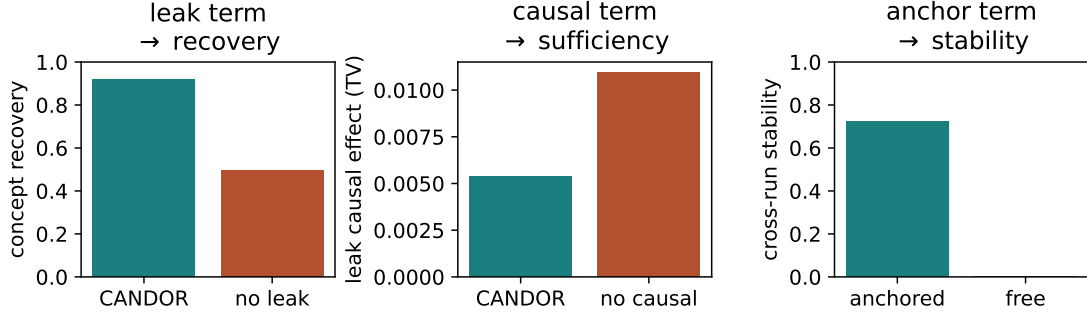


Figure 3: Objective ablations: each term is load-bearing for a *distinct* guarantee. **Left:** the leak-energy term governs concept recovery (dropping it halves recovery). **Centre:** the causal term governs causal sufficiency (dropping it roughly doubles the leak’s causal effect, mimicry without sufficiency). **Right:** the anchoring term governs cross-run concept-identity stability (dropping it sends it to zero). No single objective delivers all three; the conjunction does.

Locatello et al., 2019]. CANDOR is a concept bottleneck hardened with the guarantees this line proved necessary; *leakage is dark computation under another name*, and the certificate is what bounds it.

Certification, oversight, evaluation, governance. [Katz et al., 2017, Zhang et al., 2018, Xu et al., 2020, Przybysz et al., 2023, Ignatiev et al., 2019, Christiano et al., 2021, Roger et al., 2023, Lanham et al., 2023, Chen et al., 2025, Baker et al., 2025, Korbak et al., 2025, Jacovi and Goldberg, 2020, Karvonen et al., 2025, Amodei, 2025, Clymer et al., 2024, European Union, 2024]. CANDOR borrows the sound-over-approximation form and the safety-case framing, and supplies the runtime certificate that monitoring and CoT approaches cannot. The runtime-certificate stance also continues a line of *certifying* runtime guardrails for deployed models, but moves the certificate from the model’s *outputs* to the model’s own *computation*.

9 Limitations and honest scope

- **Scale, and the retrofit/by-construction gap.** My positive evidence is on synthetic ground-truth tasks and a one-layer Transformer, *not* a frontier LLM; I chose the regime where soundness can be *checked*. On the real pretrained GPT-2 (§7.4) the certificate *machinery* transfers, but a *frozen* retrofit does not favour CANDOR over a reconstruction-only SAE, because CANDOR’s gains require the model to reshape its computation through the channel, which only *by-construction* training provides. The single most important next experiment is therefore by-construction training (or layer-wise fine-tuning) of a pretrained LLM with the bottleneck, followed by a frontier-scale evaluation (SAEBench, RAVEL, MIB, and a real backdoor-detection battery [Karvonen et al., 2025, Huang et al., 2024, Mueller et al., 2025]); both are future work, not present claims. Whether the favourable tax/legibility frontier persists at scale is the central open empirical question.
- **No completeness.** By Rem. 1 I do not and cannot claim the certificate reaches zero on arbitrary computation; I claim it is sound, measured, and trainable, and that a large δ is an honest abstention signal.
- **“Certificate” $\neq$ formal proof.** $\delta(x)$ is a sound *per-input measurement*, not a verified bound over all inputs. It is one-sided (cannot under-report divergence for the inputs evaluated) but

says nothing, on its own, about inputs not run. A periodic exact certificate on a tractable channel fragment [Przybysz et al., 2023] and a worst-case bound-propagation audit [Xu et al., 2020] are the natural complements.

- **Anchoring tension.** Anchoring injects inductive bias to stabilise identity; pushed too hard it makes concepts the designer’s labels rather than the model’s. I report where CANDOR sits via the cross-run stability number rather than assuming the best case.
- **Concept placement.** On the planted task I place the concept layer where the generative factors are linearly present; identifying such sites automatically in a deep agent is unsolved, and computation that is irreducibly distributed across many sites will show up as leak, not as a clean concept.

10 Broader impact

This work is about transparency and oversight of capable AI. A runtime, sound, checkable bound on a model’s non-legible computation is exactly the kind of artefact a safety case for an autonomous agent needs [Clymer et al., 2024, Amodei, 2025] and that transparency regulation increasingly asks for [European Union, 2024]. The same mechanism is dual-use only in the weak sense that any interpretability tool is: it makes models easier to understand, audit, and correct. The honest risk I flag is *false assurance*: treating a low δ on a test distribution as a guarantee off-distribution. I therefore frame δ as a measurement with stated assumptions, demonstrate that it *rises* on bypassing computation rather than staying silent, and release everything needed to recompute it.

11 Conclusion

Interpretability has been practised as archaeology: build an opaque artefact, then dig. I have argued that the way toward interpretable AGI is to stop digging, to make legibility an architectural invariant and emit the audit at inference time. The Legible Bottleneck is the single primitive that makes this possible: route a layer’s computation through a sparse, typed, anchored concept code, keep and *measure* the unexplained leak, and train the concepts to be causally sufficient. The resulting CANDOR architecture certifies each of its decisions with a sound, re-checkable bound on its own dark computation, recovers ground-truth concepts at near-zero tax, and surfaces a channel-bypassing backdoor that no post-hoc method bounds. It is not a solution to interpretable AGI, and it does not claim completeness. It is a concrete, checkable mechanism for turning interpretability from a description one hopes is true into an invariant the model proves, one forward pass at a time.

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